

Control TV With Alexa IoT TV Controller

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In a smart home electronic appliances can be controlled through voice or gesture, including a TV set. Unfortunately, this feature is not available in most TVs. To use such a feature, a Fire TV stick is needed, but it gives only limited TV access.

So, here is a project to make an Alexa IoT TV controller that will solve the problem. The system works by connecting Alexa to an IoT device over Wi-Fi. The IoT device receives commands from Alexa to trigger the IR LED for controlling the set-top box or TV.

The Alexa IoT TV controller is not just restricted to controlling TVs. You can use it to control any device that uses an IR remote control. Sounds interesting? Gather your tools and let us get started.

To install the ESP8266 board in Arduino IDE, go to Files→Preferences→Paste the URL http://arduino.esp8266.com/stable/package_esp8266com_index.json. Then go to Tools→Board Manager→Search esp8266 and install the board in Arduino IDE.

Also, install the library espalexa in Arduino IDE, which will allow Alexa to control the IoT appliances that have been added to the Alexa app. Next, install the IR remote library that will allow you to decode the hex code for transmission to the TV.

Coding

We need to write two codes: one for ESP8266-01 that will be used as an



Fig. 1: Author's prototype

```
/*
 * This is a basic example on how to use Espalexa and its device declaration methods.
 */
#ifdef ARDUINO_ARCH_ESP32
#include <WiFi.h>
#else
#include <ESP8266WiFi.h>
#endif
#include <Espalexa.h>

// prototypes
boolean connectWifi();

//callback functions
void firstLightChanged(uint8_t brightness);
void secondLightChanged(uint8_t brightness);
void thirdLightChanged(uint8_t brightness);
void forthLightChanged(uint8_t brightness);
void fifthLightChanged(uint8_t brightness);
void sixthLightChanged(uint8_t brightness);
void seventhLightChanged(uint8_t brightness);
void eighthLightChanged(uint8_t brightness);

// Change this!!
const char* ssid = "IRONMAN";
const char* password = "1234567890";

boolean wifiConnected = false;
```

Fig. 2: Code for ESP8266

IoT device for Alexa and the other for Arduino that will trigger the IR LED to control the TV. For controlling the TV using Alexa, create the code using espalexa library. It also allows the addition of lighting devices to Alexa.

However, do remember to set the function for passing the control commands to Arduino for controlling the TV instead of lights.

Esp8266-01 code. For writing the code, first initialise espalexa library

Materials

Table 1

Component name	Quantity	Description
Esp8266-01	1	IoT chip
Alexa Device	1	Alexa echo
FTDI Programmer	1	Programmer module
IR LED	1	IR LED
Arduino Nano, Micro	1	For programming
3.3-5V DC Adaptor	1	DC adaptor